

- Q1) Satisfiability
- a) Truth table $\rightarrow$ 25
 tautology $\rightarrow$ 05
 no truth table $\rightarrow$ 0
- b) Truth table $\rightarrow$ 1
 tautology $\rightarrow$ 05
 modulus ponens $\rightarrow$ 05
 no truth table $\rightarrow$ 0

- c) CNF $\rightarrow$
 $(\neg P \vee Q \vee R) \wedge (P \vee \neg Q \wedge R) \wedge (P \vee R) \wedge (R \vee \neg P)$
 with steps $\rightarrow$ 2

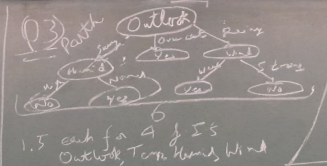
- DNF
 $(P \wedge S) \vee (P \wedge Q) \vee (P \wedge R) \vee (P \wedge R) \vee$
 $(T \wedge T \wedge R \wedge S) \vee (T \wedge T \wedge R \wedge T)$
 with steps $\rightarrow$ 2

Q2 abc: Anurag

- a) $X \rightarrow$ original bit
 $Y \rightarrow$ read bit
- $P(X=1|Y=1) = \frac{P(Y=1|X=1)P(X=1)}{P(Y=1)}$ 05
- $P(Y=1) = P(X=1|Y=1)P(X=1) + P(X=0|Y=1)P(X=0)$ 05
- $P(Y=1|X=1) = \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4} \quad P(Y=1|X=0) = \frac{1}{4}$ 05
- $P(Y=1) = \frac{3}{4} + \frac{1}{4} \cdot \frac{1}{2} \rightarrow$ 05
- Ans $\rightarrow \frac{3P}{2P+1}$ 05
 Wrong ans $\rightarrow$ 05

- b) Independent given M $\rightarrow P(SH|M) = P(S|M)P(H|M)$
- $P(M|SH) = \frac{P(SH|M)P(M)}{P(SH)} = \frac{P(S|M)P(H|M)}{P(SH)}$ 1
- c) i) $P(H=1|T=1) = \frac{P(T=1|H=1)P(H=1)}{P(T=1)}$ 1
- ii) Accuracy $\rightarrow$ 0.75 05
 Precision $\rightarrow$ 0.5 1
 Recall $\rightarrow$ 0.75 1

Q2 d) Parth
 Dist. Calc $\rightarrow$ 0
 Bias $\rightarrow$ 0
 KNN labels + pt. $\rightarrow$ 0



- 1.5 each for 4 of 5's
 Outlook, Temp, Humid, Wind
- (5) 1) Correct ans for 10 out = Bias 0, Var = 0.0022, MSE = 0.0022
 for 20 out Bias 0, Var = 0.00387, MSE = 0.00387
- 2) if weighted mean method used (final marks $\times 0.9$)
 for 10 out Bias = -0.1, Var = 2.92, MSE = 2.92
 for 20 out Bias = 0, Var = 2.93, MSE = 2.93
- if combined both sets (final marks $\times 1.6$)
 Bias = 0, Var = 0.00147, MSE = 0.00147
- if Bernoulli method used for var $\rightarrow$ full marks
 for 10 out Bias = 0, Var = 0.01366, MSE = 0.01366
 for 20 out Bias = 0, Var = 0.0067, MSE = 0.0067
- for each correct ans +1 for Bias +1 for Var +1 for MSE with brief steps showing
 if method correct, but calculation mistake $\rightarrow$ 0.75 each
 if partially correct concept $\rightarrow$ 0.5
 if only formulas written $\rightarrow$ 0.25 each
 +0.25 for finding probability of each 10 & 20 outcome
 if weighted mean approach used, or calculation mistakes
 no marks for probability calculations
 if correct concept, but finally did not divide by 6
 for combined approach calculated for 3 marks & scaled by 1.6
 for weighted means approach $\rightarrow$ final ans $\times 0.9$
 Go through key before approaching TA. If unreasonable queries, you will lose marks

- Q4 Memory
- a) Support = 1/2 0.66 marks
 Confidence = 3/4 0.66 marks
 Lift = 9/8 0.66 marks
- b) Any answer which describes that ϵ greedy enables exploration - 0.5 mark
 Mentioning "exploration" - 0.5 mark
- c) $\epsilon/2$ - 0.5 mark
 Any valid reasoning - 0.5 mark
- d) Option b - 1 mark
- e) 1, 2, 3, 1, 2, 2, 2, 3, 1, 3
 0.6 marks if these rounded. No marks if rounded, then has not been performed in first 3 rounds. Otherwise marks given by case to case basis

Q7 ~ Krishiv

@ T/F $\rightarrow 0.5$

Explanation - 1.5

Answer (b) - 1

(b) T/F - 0.5

Risk neutral modelling - 1 (0.5 given if just explained about it)

For explanation & computation - 1.5

(c) T/F - 0.5

Risk averse modelling - 1

Expⁿ and compⁿ - 1.5

(d) (0.52, 0.48) - 1 mark

Final Ans - 1 mark

0.5 - Utility FN

0.5 - Weight bet

Q6 A) 70 100
4 [2M Optimal [70 70 70 70] [100 100 100 100]
2M ML [70 70 70 70] [100 100 100 100]

4 [2B) Optimal [40.43, 40.38, 59.02, 70] [61.23, 72.52, 85.32, 100]
ML [41.07, 15.39, 59.02, 70] [21.43, 27.18, 85.33, 100]

c) 1.5 + 1.5 for calc MSE for both parts

2 Mthds of Solving. Assume $[u_1^*, u_2^*, u_3^*, 70]$ & write eqn $\rightarrow$ 4 Equation Solving

② Value iteration from $[0, 0, 0, 70]$

$\frac{1}{2}$ marks of each correct iteration (at Max for 2 iterations)

A) Optimal $\left. \begin{array}{l} v_1: [0 \ 0 \ 56 \ 70] \\ v_2: [0 \ 44.8 \ 67.2 \ 70] \end{array} \right\} \frac{1}{2}$ and so on.

ML: $\left. \begin{array}{l} v_1: [0 \ 0 \ 56 \ 70] \\ v_2: [0 \ 5.6 \ 67.2 \ 70] \end{array} \right\} \frac{1}{2}$

Q6 ABC MANAS

Q8) Jai

a) Obj w/o R — ①

Explain — ①

Obj with R — ①

Explain — ①

b) FE — modify/improve features — ①

Scaling — Scale for fixed range

Ext any standard technique — 0.5

Binning — Continuous fts. for discrete bins — ①

Q6d) Jai

Dim = 4×13 — 0.5

S_2 no. of col. — 0.5

0.25 / col.

Extra col. — -0.5

Q10) Tathya

Alt 1: Manhattan

$h(n) = 2(4 - r) + 2(c - r)$

Alt 2: h_{slo}

Alt 3: $h(n) = h(\text{true cost})$

or any other heuristic fcn

(x, y) or (r, c) to (x_0, y_0) or (r_0, c_0)

For showing $h(n) \leq h^*(n)$

Consistency All 4 cases

b) Alt 1: Manhattan, has 13 nodes

Alt 2: h_{slo} has 10-12 nodes

Alt 3: $h(n)$ has 6 nodes

1 mark for correct no. of nodes

2 marks for steps correct g, h, h(n) values

c) GBFS has around 6 nodes — ①

Correct h(n) values in steps — ①

d) GBFS has 6 while A^* has

more nodes expansion but GBFS

doesn't guarantee optimal soln

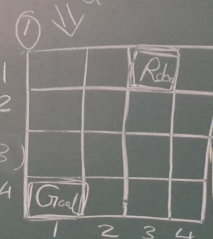
'is greedy algo & A^* is optimally efficient

the best algo among all other algo that

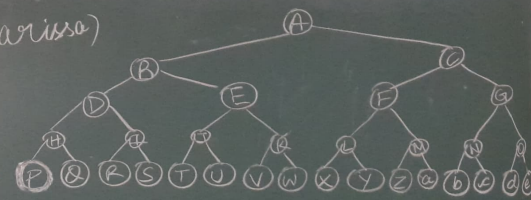
can find optimal soln cuz its consistent &

admissible → 0.5

Grid



9) Harissa



DFS stack: $A \rightarrow CB \rightarrow CED \rightarrow CEIH \rightarrow CEIUF$ (1 mark)

BFS queue: $A \rightarrow CB \rightarrow EDC \rightarrow GFED \rightarrow \dots$ (1 mark)

$\Rightarrow$ smm-BFS = 16 (1 mark)

smmBFS - smmDFS = $16 - 5 = 11$ (0.5 marks)

Reasoning (1.5 marks):

The chosen goal node gives maximum value

since smm BFS depends on width of level of

node and smm DFS depends on depth.

smmBFS: exponential increase on going down

smmDFS: linear increase on going down

Goal node. Any in last level

→ 0.5